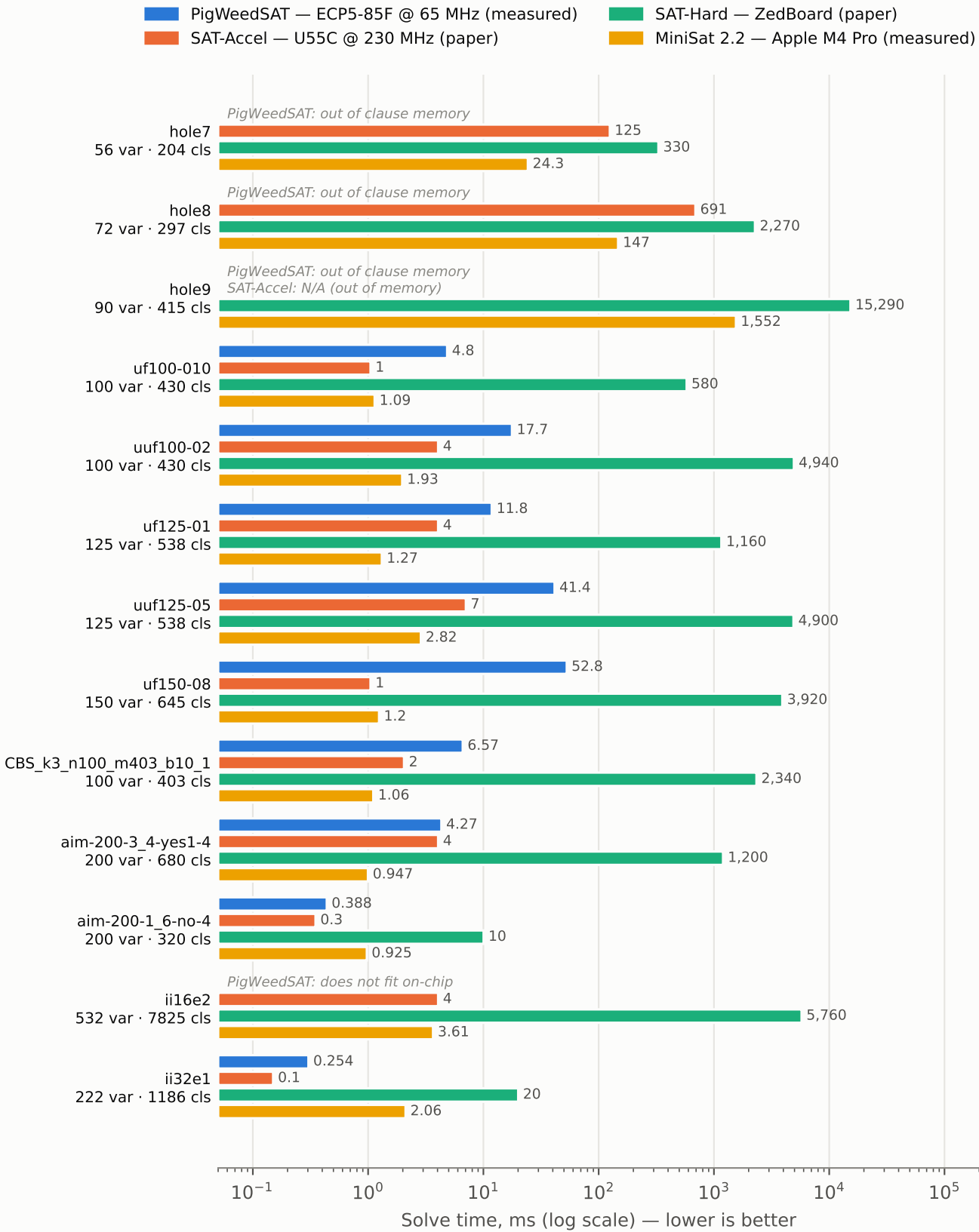




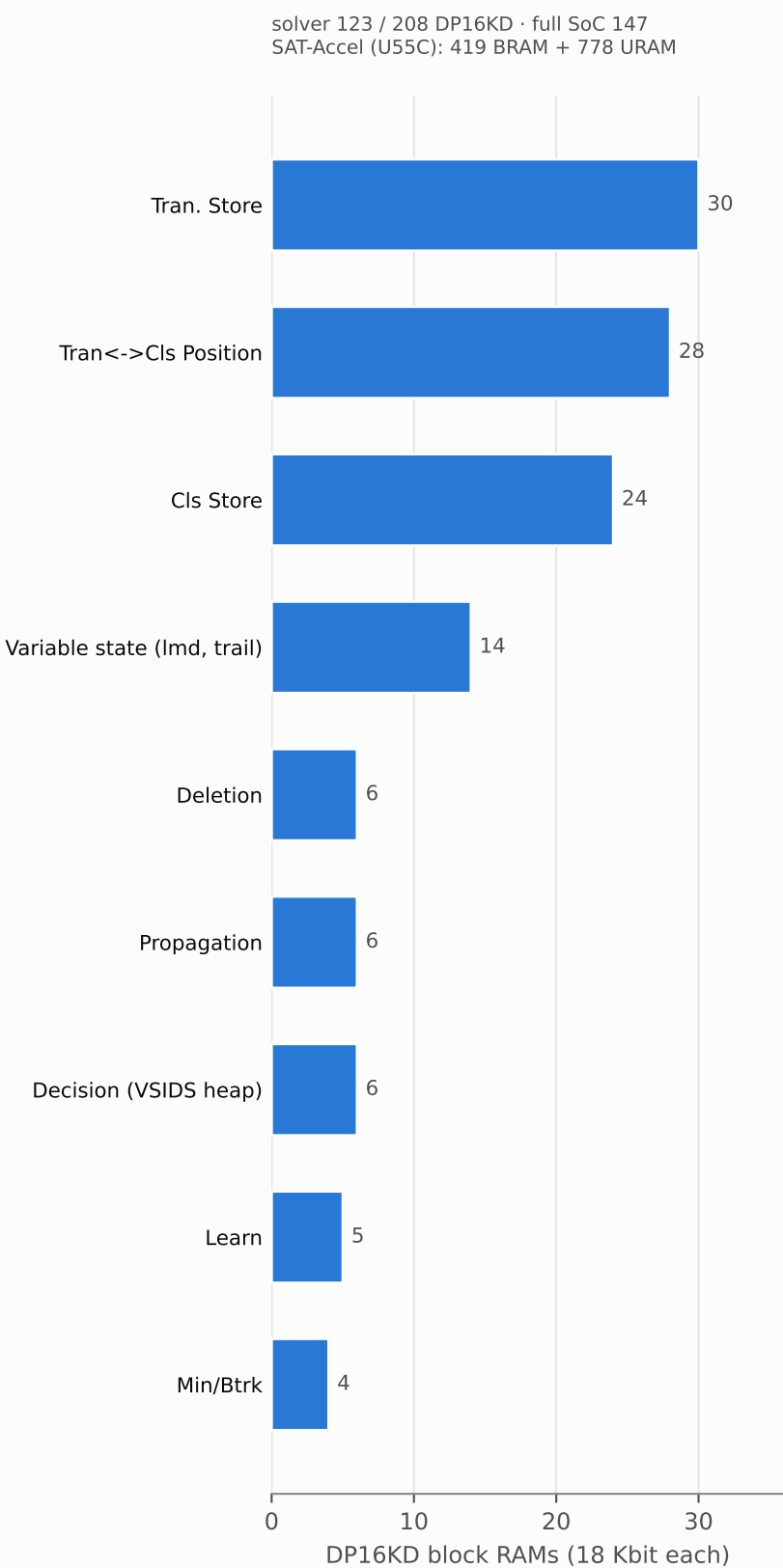
From the seed of SAT-Accel grows PigWeed (SAT).

An Amaranth HDL implementation of SAT-Accel (Lo, Chang & Cong, FPGA '25) on a Lattice ECP5-85F, measured on Manhattan Reasoning cloud FPGAs.

(a) Table 3 — SATLIB instances used by SAT-Hard



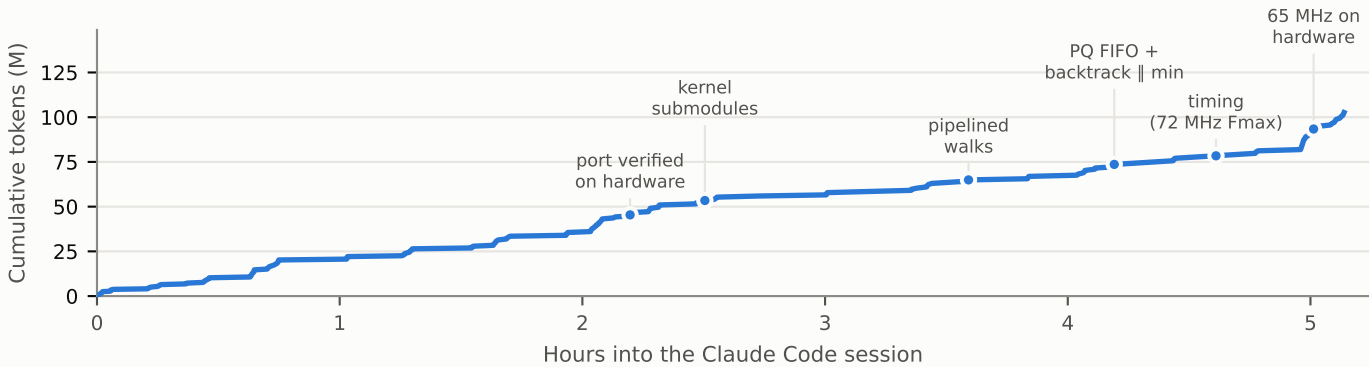
(b) Table 2 — block RAM by module



(c) Platforms

	Board	FPGA	Node	Clock	LUTs	FFs	Block RAM	DSPs	Board DRAM	Board cost
PigWeedSAT (this work)	Lattice ECP5-5G Eval	ECP5 LFE5UM5G-85F	40 nm	65 MHz	84K	84K	208 × 18 Kb (3.7 Mb)	156	—	~\$140
SAT-Accel (FPGA '25)	AMD Alveo U55C	Virtex US+ XCU55P	16 nm	230 MHz	1.30M	2.61M	2,016 × 36 Kb + 960 URAM	9,024	16 GB HBM2	~\$5,400
SAT-Hard (DSD '19)	Digilent ZedBoard	Zynq-7000 XC7Z020	28 nm	n/r	53K	106K	140 × 36 Kb (4.9 Mb)	220	512 MB DDR3	~\$620

(d) Compute spend — tokens used to build and measure PigWeedSAT



- 102.6M tokens in total
- 101.5M cache reads (99%): context re-read each turn
- 0.75M cache writes (new context)
- 0.38M output tokens (code, reasoning, tool calls)
- 194 model responses · claude-opus-5-5

PigWeedSAT: on-chip solve cycles at 65 MHz on a Manhattan Reasoning cloud ECP5-85F, host load excluded; all 9 solved answers correct and SAT models re-verified. Geometric mean over those 9: 122× faster than SAT-Hard, 3.9× slower than SAT-Accel, 4.1× slower than MiniSat on the M4 Pro. SAT-Accel and SAT-Hard times from Lo et al., FPGA '25, Table 3; SAT-Hard platform from Ustaoglu et al., DSD '19 (clock not reported, n/r). MiniSat 2.2: default options, median CPU time of 11 runs (includes parsing; ~1 ms floor). Board cost: mean of single-unit US list prices (Oct 2026), board only, no host. Compute spend: Claude Code session transcripts (tools/token_log.py); tokens as reported by the API, not dollars.